EXAM PAPERS PRACTICE

Boost your performance and confidence with these topic-based exam questions

Practice questions created by actual examiners and assessment experts

Detailed mark scheme

Suitable for all boards

Designed to test your ability and thoroughly prepare you

AQA A Level Further Mathematics (7367)

Topic	D: Further Algebra and Functions
Sub topic	D13: Graph Rationals with Quadratics

Mark Scheme

AQA A Level Further Mathematics (7367)

Mark Scheme

Topic	**D: Further Algebra and Functions**
Sub topic	**D13: Graph Rationals with Quadratics**

To be used by all students and teachers for AQA A Level Further Mathematics (7367) for exams from 2025 onwards.

Students and teachers of other boards may also find this useful

Mark schemes

Q1.

	Marking Instructions	AO	Marks	Typical Solution
(a)(i)	Eliminates y	AO1.1a	M1	$k = \dfrac{5x^2 - 12x + 12}{x^2 + 4x - 4}$
	Obtains a quadratic equation in the form $Ax^2 + Bx + C = 0$, PI by later work	AO3.1a	M1	$k(x^2 + 4x - 4) = 5x^2 - 12x + 12$ $(k - 5)x^2 + 4(k + 3)x - 4(k + 3) = 0$ (A)
	Obtains $b^2 - 4ac$ in terms of k for 'their' quadratic FT 'their' quadratic provided first M1 awarded	AO1.1b	A1F	$y = k$ intersects C_1 so roots of (A) are real $b^2 - 4ac =$ $[4(k + 3)]^2 - 4(k - 5)(-4(k + 3))$
	Obtains inequality, including > 0, where k is the only unknown for 'their' discriminant FT 'their' discriminant provided both M1 marks have been awarded	AO1.1a	M1	$16(k + 3)^2 + 16(k - 5)(k + 3) > 0$ $16(k + 3)(k + 3 + k - 5) > 0$
	Completes a rigorous argument to show that $(k + 3)(k - 1) > 0$ This mark is available only if all previous marks have been awarded	AO2.1	R1	$\Rightarrow (k + 3)(2k - 2) > 0$ $\Rightarrow (k + 3)(k - 1) > 0$
(ii)	Obtains critical values	AO1.1b	B1	Critical values are -3 and 1
	Deduces that $k = -3$ for maximum	AO2.2a	R1	$k \le -3$ (or $k \ge 1$) satisfy inequality
	Substitutes for k into 'their' quadratic from **(a)(i)** FT 'their' quadratic only if first M1 awarded in **(a)(i)**	AO1.1a	M1	For max pt, $k = -3$ Sub $k = -3$ in (A) gives $-8x^2 = 0$
	States coordinates of max pt NMS 0/4 Must be using **(a)(i)**	AO1.1b	A1 CAO	Max pt of C_1 is $(0,-3)$

(b)	Uses discriminant to determine solution	AO2.4	E1	$(-12)^2 - 4(5)(12) < 0$ $k \neq 0$ Denominator, $5x^2 - 12x + 12$ of $\frac{1}{f(x)}$ is never 0 so C_2 has no vertical asymptotes.
	Deduces no vertical asymptotes with clear reasoning with reference to denominator	AO2.2a	R1	
(c)	Obtains $y = 1$	AO3.2a	B1	$y = 1$
				Total 12 marks

Q2.

(a) (HA) $y = 1$

$y = 1$ OE eqn

B1

(VA) $x^2 - 2x - 3 = 0$ $(x - 3)(x + 1) = 0$

PI OE eg use of quadratic formula

M1

$x = -1$ and $x = 3$

Both needed OE eqn(s)

A1

3

(b) (i) $k = \dfrac{x^2 - 2x + 1}{x^2 - 2x - 3} \Rightarrow kx^2 - 2kx - 3k = x^2 - 2x + 1$

$kx^2 - 2kx - 3k - x^2 + 2x - 1 = 0$

$(k - 1)x^2 - 2(k - 1)x - (3k + 1) = 0$

AG Must see the two stages, correct elimination of fraction and a correct rearrangement to... = 0, along with correct elimination of brackets before printed answer is stated.

B1

1

(ii) Discriminant $b^2 - 4ac$ $\{ 4(k - 1)^2 + 4(k - 1)(1 + 3k) \}$

$b^2 - 4ac$, OE, in terms of k; condoning one minor error in substitution.

M1

Line intersects curve $\Rightarrow b^2 - 4ac \geq 0$
$\Rightarrow 4(k - 1)^2 + 4(k - 1)(1 + 3k) \geq 0$

A correct inequality where k is the only unknown

A1

$\Rightarrow 4(k - 1)[k - 1 + 1 + 3k] \geq 0$, $16k(k - 1) \geq 0$

ie $k^2 - k \geq 0$

CSO AG Must be convinced

A1

3

(iii) $k^2 - k \geq 0$, $k(k - 1) \geq 0$,

$k \leq 0$, $k \geq 1$ Critical values $k = 0$, ($k = 1$)

For $k = 0$ either as an equation or inequality.

B1

$k \neq 1$ since there is no point on the curve where $y = 1$ ($x^2 - 2x - 3 \neq x^2 - 2x + 1$)

OE Valid explanation, with no accuracy errors, to discount $k = 1$

E1

$k = 0$, $-x^2 + 2x - 1 = 0$ or $y = 0$, $x^2 - 2x + 1 = 0$

OE

M1

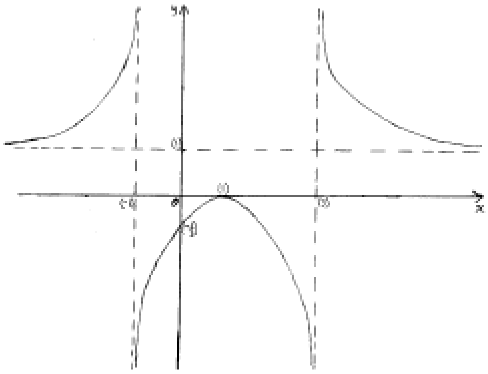
(Only one) **stationary** point (and its coordinates are) **(1, 0)**

'stationary' with either (1, 0) or {x = 1, y = 0}

A1

4

(c)



Curve with three distinct branches

B1

Branch between VAs, correct shape, no part of the branch above the x-axis, only intersection with y-axis at a point below the origin, and its max pt on the positive x-axis

B1

Fully correct curve drawn with each branch correctly approaching its relevant asymptotes

B1

3

[14]

Q3.

(a) Asymptotes

$x = -1$

$x = -1$ OE

B1

$x = 2$

$x = 2$ OE

B1

$y = 0$

$y = 0$

B1

3

(b) $-\frac{1}{2} = \frac{x}{x^2 - x - 2} \Rightarrow x^2 - x - 2 = -2x$

Correctly removing brackets and fractions to reach $x^2 - x - 2 = -2x$ OE

M1

$x^2 + x - 2 = 0 \Rightarrow x = 1, x = -2$

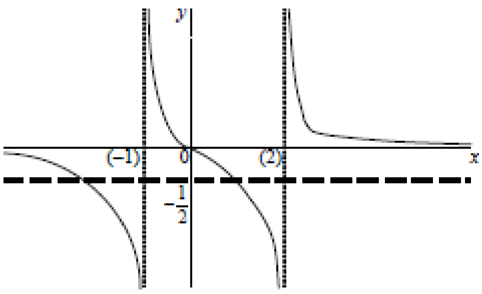
Correct two values for x-coordinates.

NMS 2 or 0 marks

A1

2

(c)



Three branches shown on sketch of C with either middle branch or outer two branches correct in shape

M1

All three branches, correct shape and positions and approaching correct asymptotes in a correct manner.

*If middle branch does **clearly** not go through the origin, then A0*

A1

Correct sketch of line (L), $y = -0.5$ identified

B1

3

(d) $-2 \le x < -1$

Condone

B1

$1 \le x < 2$

Condone

B1

$-2 \le x < -1$, $1 \le x < 2$

All complete and correct

B1

3

[11]

Q4.

(a) Denom never zero, so no vert asymp

E1

Horizontal asymptote is $y = 0$

B1

2

(b) $x - 4 = k(x^2 + 9)$

M1

Hence result clearly shown

AG

A1

2

(c) Real roots if $b^2 - 4ac \ge 0$

PI (at any stage)

E1

Discriminant = 1 − 4k(9k + 4)

M1

... = −(36k^2 + 16k − 1)

m1 for expansion

m1

... = −(18k − 1)(2k + 1)

m1 for correct factorisation

m1

Result (AG) clearly justified

eg by sketch or sign diagram

A1

5

(d) $k = -\frac{1}{2} \Rightarrow -\frac{1}{2}x^2 - x - \frac{1}{2} = 0$

or equivalent using $k = \frac{1}{18}$

M1A1

... $\Rightarrow (x+1)^2 = 0 \Rightarrow x = -1$

A1

$k = \frac{1}{18} \Rightarrow \frac{1}{18}x^2 - x + \frac{9}{2} = 0$

A1

... $\Rightarrow (x-9)^2 = 0 \Rightarrow x = 9$

A1

SPs are $\left(-1, -\frac{1}{2}\right), \left(9, \frac{1}{18}\right)$

correctly paired

A1

6

[15]

Q5.

(a) Asymptotes x = 2, y = 0

B1B1 **2**

(b) One correct branch

B1

Both branches correct

no extra branches; $x = 2$ shown

B1 **2**

[4]

Q6.

(a) (i) Asymptotes $x = 0$, $x = 2$, $y = 1$

B1 × 3 **3**

(ii) Intersections at (1, 0) and (3, 0)

B1 **1**

(iii) At least one branch approaching asymptotes

B1

Each branch

B1 × 3 **4**

(b) $0 < x < 1$, $2 < x < 3$

Allow B1 if one repeated error occurs, eg ≤ for <

B1,B1 **2**

Alternative:
Complete correct algebraic method

M1A1 **(2)**

[10]

Q7.

(a) Asymptotes $x = 1$, $x = 5$, $y = 1$

B1 × 3 **3**

(b) $y = -1 \Rightarrow (x - 1)(x - 5) = -x^2$

OE

M1

... $\Rightarrow 2x^2 - 6x + 5 = 0$

OE

m1

Disc't = 36 – 40 < 0, so no pt of int'n

convincingly shown (AG)

A1

3

(c) (i) $y = k \Rightarrow x^2 = k(x^2 - 6x + 5)$

OE

M1

... $\Rightarrow (k - 1)x^2 - 6kx + 5k = 0$

convincingly shown (AG)

A1

2

(ii) Discriminant = $36k^2 - 20k(k - 1)$

OE

M1

... = 0 when $k(4k + 5) = 0$

convincingly shown (AG)

A1

2

(d) $k = 0$ gives $x = 0$, $y = 0$

B1

$k = -\frac{5}{4}$ gives $-\frac{9}{4}x^2 + \frac{30}{4}x - \frac{25}{4} = 0$

OE

M1A1

$(3x - 5)^2 = 0$, so $x = \frac{5}{3}$

A1

$y = -\frac{5}{4}$

B1

5

[15]

Q8.

(a) Asymptotes $x = 0$, $x = 4$, $y = 0$

B1 × 3

3

(b) $y = k \Rightarrow 2 = kx(x - 4)$

M1

... $\Rightarrow 0 = kx^2 - 4kx - 2$

A1

Discriminant = $(4k)^2 + 8k$

m1

At SP $y = -\frac{1}{2}$

not just $k = -\frac{1}{2}$

A1

... $\Rightarrow 0 = -\frac{1}{2}x^2 + 2x - 2$

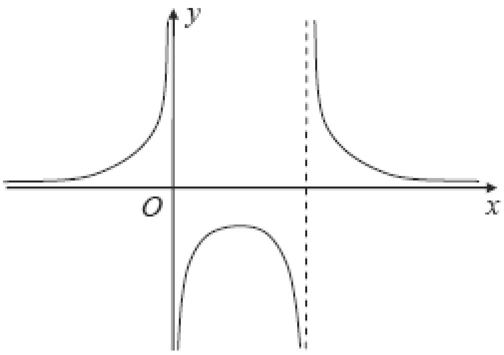
m1

So $x = 2$

A1

6

(c)



Curve with three branches approaching vertical asymptotes correctly

B1

Outer branches correct

B1

Middle branch correct

B1

3

[12]

Q9.

(a) Asymptotes $y = 0$, $x = -1$, $x = 1$

B1 × 3

3

(b) Three branches approaching two vertical asymptotes

Asymptotes not necessarily drawn

B1

Middle branch passing through O

with no stationary points

B1

Curve approaching $y = 0$ as $x \to \pm\infty$

B1

All correct

with asymptotes shown and curve approaching all asymptotes correctly

B1

4

(c) Critical values $x = -1$, 0 and 1

B1

Solution set $-1 < x < 0$, $x > 1$

M1 if one part correct
or consistent with c's graph

M1A1

3

[10]

Q10.

(a) (i) Intersections at (−1, 0), (3, 0)

Allow x = −1, x = 3

B1B1

2

(ii) Asymptotes $x = 0$, $x = 2$, $y = 1$

B1 × 3

3

(b) (i) $y = k \Rightarrow kx^2 - 2kx = x^2 - 2x - 3$

M1 for clearing denominator

M1A1

... $\Rightarrow (k - 1)x^2 + (-2k + 2)x + 3 = 0$

ft numerical error

A1ft

$\Delta = 4(k - 1)(k - 4)$, hence result

convincingly shown (AG)

m1A1

5

(ii) $y = 4$ at SP

B1

$3x^2 - 6x + 3 = 0$, so $x = 1$

A0 if other point(s) given

M1A1

3

(c) Curve with three branches

approaching vertical asymptotes

B1

Middle branch correct

coordinates of SP not needed

B1

Other two branches correct

3 asymptotes shown

B1

3

[16]

Q11.

(a) Asymptotes $x = 0$, $y = 1$

B1, B1

2

(b) (i) $\Delta = 4 - 8 < 0$, so num never 0

OE; E1 for incomplete explanation

E2,1

2

(ii) Method for solving quadratic

"i" must appear

M1

Roots –1 ± i

A1 if one error made

A2,1

3

(c) (i) $\mathrm{f}(x) = k \Rightarrow x^2 + 2x + 2 = kx^2$

M1

$\ldots \Rightarrow (1-k)x^2 + 2x + 2 = 0$

m1

Equal roots $\Rightarrow 4 - 8(1-k) = 0$

Convincingly shown (AG)

A1

3

(ii) $k = \frac{1}{2}$

B1

$y = \frac{1}{2}$ ***at SP***

ft wrong value for k

B1ft

So $\frac{1}{2}x^2 + 2x + 2 = 0$

M1

and $x = -2$ at SP

A1

4

[14]

Q12.

(a) (i) Asymptote is $y = 1$

B1 **1**

(ii) Denominator never zero

E1 if incomplete

E2,1 **2**

(b) $f(x) = k \Rightarrow (1 - k)x^2 + 4x - 9k = 0$

M1A1

Equal roots if $16 + 36k(1 - k) = 0$

m1

ie if $9k^2 - 9k - 4 = 0$

convincingly shown (AG)

A1 **4**

(c) Solving quadratic for k

M1

$k = -\frac{1}{3}$ or $k = \frac{4}{3}$

NMS 2/2

A1

$4x^2 + 12x + 9 = 0$ or $x^2 - 12x + 36 = 0$

m1

SPs when $x = -\frac{3}{2}$ ….; and when $x = 6$…….

A1A1

SPs are $\left(-\frac{3}{2}, -\frac{1}{3}\right)$ and $\left(6, \frac{4}{3}\right)$

A1 **6**

[13]

Examiner reports

Q2.

Students showed that they had prepared well for this style of question and some very good solutions were seen although full marks was a rarity due mainly to the demands of part (b)(iii). In part (a) many students gave the correct three equations of the asymptotes although '$y = 0$' was a common wrong answer. In part (b)(i) errors were made in rearranging the terms resulting in incorrect signs. Although the printed answer was frequently stated, it did not always follow from correct previous lines. Again, for a printed answer, the details of all steps should be shown and any alterations applied to all relevant lines of working. In part (b)(ii), the majority of students attempted to find the discriminant and then use $b^2 - 4ac \geq 0$, although miscopies of the terms, the negative sign lost in $-(3k + 1)$ and the '2' not squared in finding b^2 were all seen, sometimes followed by a further error to produce the printed answer. Solutions to part (b)(iii), when attempted, did not always address the 'Hence' or show that there was only one stationary point. Many students missed the fact that $k = 0$ was a critical value of the printed inequality and a significant number claimed that $k = 1$ produced the stationary point (1, 0). Better solutions used $k = 0$ to find the stationary point but many of these did not attempt to give an explanation of why $k = 1$ was excluded or gave the unacceptable explanation '$y = 1$ is an asymptote' to dismiss it. In part (c) the quality of the sketches was very variable. There were many good sketches seen but some other students would have found it easier to show the behaviour of the curve as it approached the asymptotes if they had used a ruler to draw the asymptotes and so had clear vertical and horizontal (dashed) lines.

Q3.

In part (a), the equations of the vertical asymptotes were usually stated correctly, but the equation $y = 0$ of the horizontal asymptote was occasionally omitted or an incorrect equation, usually $y = 1$, was given. The very few students who failed to express the asymptote in the form of an **equation** gained no credit. In part (b), students generally understood the requirements but algebraic errors in rearrangement and expansion of brackets sometimes led to incorrect quadratic equations and solutions. In part (c), the curve was generally well sketched showing the main features of the three branches. The asymptotes were usually included which significantly reduced the number of times a mark was lost due to overlapping branches. The examiners also expected to see the central branch passing through the origin which was not always the case in students' sketches. The behaviour of the curve as it approached asymptotes was generally appropriate. The line $y = -1 / 2$ was not always clearly defined either by scale or label and surprisingly even oblique lines were occasionally seen.

In part (d), average and better students often gained two marks but some lost the third mark because they did not use strict inequality signs at the –1 and 2 boundaries. A large proportion of students found this final part challenging and some did not use the graph they had drawn in part (c), preferring to embark on a fresh, complex, algebraic approach to solving the inequality which rarely led to success.

Q4.

Part (a) of this question was not always answered as well as expected. Many candidates gave a good explanation for the absence of a vertical asymptote but omitted to attempt the equation of the horizontal asymptote. When both parts were answered, the attempts were usually successful, but an equation $y = 1$ instead of $y = 0$ was quite common.

Part (b) was absolutely straightforward and afforded two easy marks to almost all the candidates.

Part (c) involved inequalities, and while most candidates were familiar with the need to work on the discriminant at this stage, many lost a mark through not clearly and correctly stating the condition for real roots; another mark was lost when the candidate, having legitimately obtained the two critical values, failed to justify the inequalities in the final answer. Again, a sign error in the manipulation of the discriminant often spoiled the attempt to find the two critical values.

It was good to see many candidates gaining full credit in part (d) even when they had struggled unsuccessfully in part (c). The majority of candidates had clearly practised their techniques in this type of question. Sometimes the finding of the y-values required an unwarranted amount of effort, since they were known from the outset, and some candidates lost the final mark because of a failure to give the correct y-coordinates.

Q5.

Most candidates started well by writing down $x = 2$ as the equation of one asymptote to the given curve, and then struggled to find the horizontal asymptote, though most were ultimately successful. The graph was often drawn correctly but almost equally often it appeared with one of its branches below the x-axis.

Q6.

The first two parts of this question were generally well answered, and there were many good sketches in part (a)(iii), though the middle branch of the curve was often badly drawn, many candidates appearing to ignore the helpful information provided about there being no stationary points. Part (b) could be answered very easily by looking at the parts of the graph which were below the x-axis, and reading off the two intervals in which this was the case. Instead, many candidates used a purely algebraic approach, and in most cases the inequality was rapidly reduced to the incorrect form $(x - 1)(x - 3) < 0$, though some candidates obtained the correct version, $x(x - 1)(x - 2)(x - 3) < 0$, and after making a table of signs came up with the correct solution.

Q7.

Part (a) was usually well answered, with most candidates writing down the equations of the two vertical asymptotes but then struggling to establish the equation of the horizontal asymptote; although the correct answer $y = 1$ appeared more frequently than the most common wrong answer, $y = 0$.

Part (b) was slightly unfamiliar, but most candidates tackled it confidently. A sign error often led to the disappearance of the x^2 term, while in other cases the quadratic and its discriminant were correctly found but the candidates omitted to mention the essential feature of the discriminant, ie the fact that it was negative, indicating the absence of any points of intersection.

Candidates were mostly on familiar ground with part (c), but marks were sometimes lost in part (c) (i) by a failure to show enough steps to justify the printed answer, and in part (c) (ii) by a sign error which prevented the candidate from reaching the required equation legitimately.

Most candidates were able to make a good attempt at part (d) in the time remaining for them. The origin was usually given correctly as one stationary point, and many found the x-coordinate of the other stationary point after a more or less lengthy calculation. Many failed to see that the y-coordinate was simply the value of k used in this calculation, but they were able to calculate it correctly from the equation of the curve.

Q8.

Most candidates scored well in parts (a) and (c) of this question. Many wrote down the equations of the two vertical asymptotes without any apparent difficulty, but struggled to find the horizontal asymptote, although in most cases they were successful. The sketch-graph in part (c) was usually drawn correctly.

Part (b) was a standard exercise for the more able candidates. No credit was given for asserting that because $x = 0$ and $x = 4$ were asymptotes, it followed that $x = 2$ must provide a stationary point. A carefully reasoned argument based on the symmetry of the function in the denominator would have earned credit, but most candidates quite reasonably preferred to adopt the standard approach.

Q9.

Most candidates scored well here, picking up marks in all three parts of the question. Those who failed to obtain full marks in part (a) were usually candidates who gave $y = 1$ instead of $y = 0$ as the equation of the horizontal asymptote. The sketch was usually reasonable but some candidates showed a stationary point, usually at or near the origin, despite the helpful information given in the question. There were many correct attempts at solving the inequality in part (c), though some answers bore no relation to the candidate's graph.

Q10.

Part (a) of this question allowed most candidates to earn marks, the examiners being lenient this time with regard to misuse of notation such as 'vertical asymptote = 2'. In part (b)(i) the majority of candidates were familiar with the required method, obtaining a quadratic equation in x and then considering the discriminant of this quadratic. Unfortunately many candidates made a sign error in manipulating the quadratic equation and so were unable to obtain the printed answer. It was distressing to see how often a candidate would falsely claim to have reached the answer legitimately.

Part (b)(ii) revealed much confusion in the minds of many candidates, who seemed to make a connection between 1 as a critical value of k and 1 as a value of x at the stationary point. Others went through two calculations, in effect repeating work already done earlier in the question, one leading to the conclusion that y could not be 1 on the curve, and the second leading correctly to the required point. Most attempts at the sketch of the curve in part (c) were either all correct or completely wrong. Occasionally a candidate would earn just one mark out of three for drawing a curve which approached the asymptotes in the correct way, or for drawing the middle branch correctly. It was not uncommon to see the middle branch completely omitted, presumably because it did not appear on the candidate's graphics calculator.

Q11.

Part (a) was generally well answered, although the equation $y = 0$ was often given instead of $y = 1$ for the horizontal asymptote. Attempts at part (b)(i) were often inconclusive, but part (b)(ii) was often well attempted despite errors in simplifying the expression $\frac{-2 \pm \sqrt{-4}}{2}$. Part (c) was only seriously attempted by stronger candidates, of whom many produced concise accurate solutions in both parts. No credit was given for attempts to use differentiation to find the stationary point.

Q12.

Some candidates were able to answer this question fully; others were able to pick up some marks, especially in part (a) where they showed some awareness of the properties of rational functions. In part (b) some candidates confused ‘equal roots’ with ‘real roots’, while others thought that the equation $9k^2 - 9k - 4 = 0$ was supposed to have equal roots. An approach adopted by quite a number of candidates was to solve this equation for k, obtaining two values, and then to show that the equation $f(x) = k$ had two equal roots for each of these two values of k. This was elongated but had the merit of leaving very little else to do in order to answer part (c) as well as part (b). The question was designed to test the candidates’ knowledge of the method laid down in this unit for finding the stationary points of a rational function. No credit was given to candidates who used differentiation to find the stationary points.